Module 1 slides: Introduction (worked examples)

Module 1 - Introduction & Math Review · Lecture slides, worked examples · 17 Aug 2026

Name _____ Date _____

Plasma drug levels after a dose were as follows.

Time (hr)	Plasma drug level (mcg/mL)
0.5	38.9
1	30.3
2	18.4
3	11.1
4	6.77
5	4.10

Slides 24–25

What is the area under the curve from hours 0.5 to 2?

Slides 24–25

What is the area under the curve from hours 2 to 4?

Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working, where her sheet gives no worked value.

Slides 24–25 Her key

Answer: 41.65 mcg·hr/mL

WHAT THE STEM GIVES

C at 0.5 hr	38.9 mcg/mL	the first side of segment 1
C at 1 hr	30.3 mcg/mL	the shared side of segments 1 and 2
C at 2 hr	18.4 mcg/mL	the last side of segment 2
widths	0.5 to 2 hr	segment 1 is 0.5 hr wide, segment 2 is 1 hr wide
other rows	11.1, 6.77, 4.10 mcg/mL	*not needed: outside hours 0.5 to 2*

THE WORKING, ONE LINE AT A TIME

SETUP

Hours 0.5 to 2 contain two segments of unequal width: 0.5 to 1 hr, which is 0.5 hr wide, and 1 to 2 hr, which is 1 hr wide.

The sampling times in this table are not evenly spaced, so each segment carries its own width and a single width cannot be factored out of the sum. Reading the widths off the time column before starting is what prevents the first segment being given a full hour.

ALGEBRA

Segment 1: [(38.9 + 30.3) mcg/mL ÷ 2] × 0.5 hr = 34.6 × 0.5 = 17.30 mcg·hr/mL.

The two concentrations average to 34.6 mcg/mL and the interval is half an hour, so the area is half of that average. Micrograms per millilitre multiplied by hours gives micrograms times hours per millilitre.

ALGEBRA

Segment 2: [(30.3 + 18.4) mcg/mL ÷ 2] × 1 hr = 24.35 × 1 = 24.35 mcg·hr/mL.

This segment is wider and its concentrations are lower, and the two effects work against each other, which is why it is the larger of the two areas despite sitting later on the curve. The width has to be taken from the time column rather than assumed.

ALGEBRA

AUC = 17.30 + 24.35 = 41.65 mcg·hr/mL.

The segment areas share a unit and add directly. The result is larger than the area from hours 2 to 4 because the concentrations over this earlier interval are higher, which is the expected direction for a curve that is declining throughout.

ROUND

AUC = 41.65 mcg·hr/mL.

The value is reported to two decimal places, which is what the half-hour and one-hour segment widths support without inventing precision. The unit is again a concentration multiplied by a time, unchanged by the unequal segment widths used over this earlier stretch of the curve.

DOES THE NUMBER MAKE SENSE?

> The span from 0.5 to 2 hr is one and a half hours wide, so the area must lie between one and a half times 18.4 and one and a half times 38.9 mcg·hr/mL, and 41.65 does.

Slides 24–25 Her key

Answer: 23.7 mcg·hr/mL

WHAT THE STEM GIVES

C at 2 hr	18.4 mcg/mL	the first side of segment 1
C at 3 hr	11.1 mcg/mL	the shared side of segments 1 and 2
C at 4 hr	6.77 mcg/mL	the last side of segment 2
widths	2 to 4 hr	each segment 1 hr wide: $(t_n - t_{n-1}) = 1$
other rows	38.9, 30.3, 4.10 mcg/mL	*not needed: outside hours 2 to 4*

THE WORKING, ONE LINE AT A TIME

SETUP

The trapezoidal rule takes each segment as $[\frac{C_{n-1} + C_n}{2}] \times (t_n - t_{n-1})$, which is one half the sum of the two parallel sides times the width.

Each pair of neighbouring points is joined by a straight line, making a trapezium whose parallel sides are the two concentrations and whose width is the time between them. Dr. Mosley reduces the printed formula to one half base times height. Summing the segments over the interval asked for gives the area across that interval.

SETUP

Hours 2 to 4 contain two segments: 2 to 3 hr and 3 to 4 hr, each 1 hour wide.

Only the points inside the requested interval are used, so the 0.5-hour and 1-hour readings play no part and the 5-hour reading is beyond the upper limit. Counting the segments before computing anything prevents both including a segment outside the range and dropping one inside it.

ALGEBRA

Segment 1: [(18.4 + 11.1) mcg/mL ÷ 2] × 1 hr = 14.75 × 1 = 14.75 mcg·hr/mL.

Averaging the two concentrations gives the height of a rectangle of equal area to the trapezium, and multiplying by the one-hour width gives the area. Micrograms per millilitre multiplied by hours gives micrograms times hours per millilitre.

ALGEBRA

Segment 2: [(11.1 + 6.77) mcg/mL ÷ 2] × 1 hr = 8.935 × 1 = 8.935 mcg·hr/mL.

The same construction applies to the second trapezium, with the 3-hour concentration now the taller of its two sides. This segment is smaller than the first because the concentration has fallen further.

ALGEBRA

AUC = 14.75 + 8.935 = 23.685 mcg·hr/mL.

The two segment areas are in the same unit and add directly, because each was already reduced to a concentration multiplied by a time. Adding rather than averaging is what the trapezoidal rule requires, since the segments sit side by side under the curve rather than overlapping.

ROUND

AUC = 23.7 mcg·hr/mL.

Dr. Mosley reports 23.7 for this interval, to three significant figures, which matches the precision of the tabulated concentrations. The unit is a concentration multiplied by a time, which is why it reads as micrograms times hours per millilitre rather than as a concentration.

DOES THE NUMBER MAKE SENSE?

The span is 2 hr wide, so the area must lie between 2 × 6.77 and 2 × 18.4 mcg·hr/mL, and 23.7 does.